

A Mathematical Introduction to Quantum Error Correcting Codes

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Abstract

Quantum error-correcting codes (QECCs) protect quantum information encoded in qubits from noise and decoherence, analogous to how classical error-correcting codes protect bits transmitted through noisy channels. This survey provides a beginner-friendly introduction to the theory of quantum error correction and several important classes of quantum codes. It assumes only basic familiarity with linear algebra, group theory, and standard quantum-mechanical notation. Foundational concepts such as quantum noise models, simple repetition-based codes, and stabilizer and CSS constructions are introduced before outlining modern code families that are actively researched today.

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1 Introduction

In 1994, Peter Shor gave an algorithm (commonly known as Shor's Algorithm [**SHOR FACTOR**]) that factorizes integers into their prime factors in polynomial time on a quantum computer- much faster than any known classical algorithm. This seemed as a threat to commonly used public key cryptography schemes, like RSA (see [**RSA**]) based on prime factorization of large numbers being slow, potentially putting communication secrecy and security at risk. However quantum systems are extremely vulnerable to errors and decoherence. Factorization of small numbers seemed unlikely, let alone the thousand-bit big integers used in cryptography.

In 1995 Shor came up with the solution. He developed Shor's Code (explained later) and the field of quantum error correction came into being. The theory of quantum error correction quickly developed, producing more specialized codes and results- some of which are explained in this survey in more detail. Now we shall now study some required theory to understand the same.

2 Quantum Information and Noise

2.1 Quantum Information

A qubit is the fundamental unit of quantum information. It is described as:-

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $\alpha, \beta \in \mathbf{C}$ such that $|\alpha|^2 + |\beta|^2 = 1$.

Multiple qubit systems also exist, for example:

$$|\psi\rangle = \alpha|001\rangle + \beta|100\rangle + \gamma|101\rangle,$$

here $|\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1$. An n qubit state can store information in a superposition of their basis states, giving access to computational space sized at most 2^n at one time. Quantum Algorithms that provide a quantum advantage/ speedup over classical algorithms are constructed using the previous fact.

However, due to practical issues like hardware problems, energy disruptions, noise in transmission channels, etc. some qubits might change creating errors. Hence it is necessary to develop a framework for studying different types of noise and perform quantum error correction for those altered qubits.

2.2 Quantum Noise

Consider a channel modifying a quantum state $|\psi\rangle$ with probability p . Its **operator sum notation** is described below:

$$N|\psi\rangle = \sqrt{1-p}E_0|\psi\rangle\langle\psi|E_0^\dagger + \sqrt{p}E_1|\psi\rangle\langle\psi|E_1^\dagger$$

Here E_0, E_1 are operation elements: E_0 is identity operator causing no change and E_1 is the operator causing change to the state.

Now, let $|\psi\rangle$ be the encoded¹ state before the noise acts. After the noise has acted the state is $\sqrt{1-p}E_0|\psi\rangle\langle\psi|E_0^\dagger + \sqrt{p}E_1|\psi\rangle\langle\psi|E_1^\dagger$. Refer section 8.2 of Chapter 8 of [1] for more details of operator sum notation. Now let us look at some examples of these types of quantum noise or errors.

2.2.1 Bit flip Channel

The bit flip channel flips the state of a qubit from $|0\rangle$ to $|1\rangle$ (and vice versa) with probability p and leaves it unchanged with probability $1-p$:

$$a|0\rangle + b|1\rangle \longrightarrow a|1\rangle + b|0\rangle$$

It has operation elements:

$$E_0 = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, E_1 = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

2.2.2 Phase Flip Channel

The phase flip channel, similar to bit flip channel, flips the phase with probability p :

$$a|0\rangle + b|1\rangle \longrightarrow a|0\rangle - b|1\rangle$$

and has operation elements

$$E_0 = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, E_1 = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

¹i.e., after applying some quantum error correcting code.

2.2.3 Bit-Phase flip channels

The bit-phase flip channel has operation elements

$$E_0 = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, E_1 = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}.$$

The name bit-phase flip is due to the fact that $Y = iXZ$, i.e., a combination of a phase flip and a bit flip operator.

3 Quantum Error Correction

Quantum errors can be digitized (i.e., by encoding the qubits such that they become equivalent to logical bits 0 and 1) and certain techniques from classical error correction are used in quantum error correction. However, the following challenges still persist, but fortunately solutions exist for all.

1. Classical codes work under the assumption that data can be arbitrarily duplicated. **The no-cloning theorem** for quantum states tells us that it is impossible to construct a cloning unitary operator U_{clone} performing:

$$U_{\text{clone}} (|\psi\rangle \otimes |0\rangle) \rightarrow |\psi\rangle \otimes |\psi\rangle$$

where $|\psi\rangle$ is the state to be cloned. Therefore it is necessary to find alternative ways of adding redundancy to the system in for quantum error correction.

2. In classical coding, only bit-flip errors need to be considered. Qubits are susceptible to both **bit-flips** (X -errors) and **phase-flips** (Z -errors). Quantum error correction codes must therefore be designed with the ability to detect both error-types simultaneously.
3. In classical error-correction we observe the output from the channel, and decide the decoding procedure to adopt on the same output. However for quantum error correction, the problem is that measurement of a **quantum state destroys its quantum information**. Hence for quantum codes, any measurements of the qubits performed as part of the error correction procedure must be carefully chosen to avoid *wavefunction collapse* and destruction of the encoded information. A special type of projective measurement referred to as a stabilizer or syndrome measurement is used to achieve this.

3.1 Simple Quantum Error Correcting Codes

Suppose a bit is to be sent from one location to another via a noisy ‘bit fli’ channel (described in Section 2.2.1). To protect the bit against effects of the noise, replace the bit with three copies of itself:

$$0 \rightarrow 000$$

$$1 \rightarrow 111$$

and send all three bits through the channel. Now $|0_L\rangle \equiv |000\rangle$ and $|1_L\rangle \equiv |111\rangle$ denote the logical zero and one states. The circuit to achieve the same is shown below.

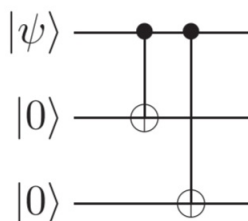


Figure1: Circuit For Creating Repetition Code States

The receiver then receives the output of the channel. If 000 was sent the output was 010 $\implies$ the second bit was flipped by the channel with probability p . It seems likely that the second bit was flipped and original bit was 000. This is called majority voting- which fails if two or more bits are flipped, causing an error with probability $p_e = 3p^2(1 - p) + p^3$. Now let us see some codes based on this principle.

3.1.1 Bit Flip Code

Consider the initial state $a|0\rangle + b|1\rangle$ encoded as $a|000\rangle + b|111\rangle$ using the procedure described in the previous section. Let a bit flip occur on at most one qubit. The following two stage error-correction procedure is carried out:

1. **Error-Detection Or Syndrome Diagnosis:** We perform a projection-operator measurement to know the type of error which has occurred. The measurement result is called the error syndrome or stabilizer. There are four error syndromes for the bit flip channel, corresponding to the four projection

operators:

$$P_0 \equiv |000\rangle\langle 000| + |111\rangle\langle 111| \text{ no error}$$

$$P_1 \equiv |100\rangle\langle 100| + |011\rangle\langle 011| \text{ bit flip on qubit one}$$

$$P_2 \equiv |010\rangle\langle 010| + |101\rangle\langle 101| \text{ bit flip on qubit two}$$

$$P_3 \equiv |001\rangle\langle 001| + |110\rangle\langle 110| \text{ bit flip on qubit three.}$$

The outcome of the measurement result (called the value of the error syndrome) is i - indicating bit flip on qubit i with the probability $\langle \psi | P_i | \psi \rangle = 1$. The syndrome measurement does not cause any change to the state. Syndrome contains only information about what error has occurred, and does not allow us to infer anything about the value of a or b .

2. **Recovery:** We use the value of the error syndrome to tell us what procedure to use to recover the initial state. The recovery procedure for each case is:

- (a) 0 (no error) - do nothing;
- (b) 1 (bit flip on first qubit) - flip the first qubit;
- (c) 2 (bit flip on second qubit) - flip the second qubit again;
- (d) 3 (bit flip on third qubit) - flip the third qubit again.

Example 3.1. *Let the bit flip occur on second qubit, so the corrupted state is $a|010\rangle + b|101\rangle$. Observe $\langle \psi | P_2 | \psi \rangle = 1 \implies$ outcome of the measurement result (the error syndrome) is certainly 2. We flip the second qubit, recovering the original state $a|000\rangle + b|111\rangle$ with perfect accuracy.*

This error-correction procedure works perfectly only when bit flips occur on one or fewer of the three qubits. This occurs with probability: $P[\text{No Change}] + P[\text{Change in any one qubit}] = (1-p)^3 + 3p(1-p)^2 = 1 - 3p^2 + 2p^3$. The probability of an error remaining uncorrected is therefore $3p^2 - 2p^3$.

3.1.2 Three Qubit Phase Flip Code

In the phase flip channel section 2.2.2, the relative phase of states $|0\rangle$ and $|1\rangle$ states is flipped with probability p , i.e., $a|0\rangle + b|1\rangle \rightarrow a|0\rangle - b|1\rangle$. Classical channels don't have any property equivalent to phase. Hence we convert the phase flip channel into a bit flip channel.

Consider the basis $|+\rangle \equiv (|0\rangle + |1\rangle)/\sqrt{2}$, $|-\rangle \equiv (|0\rangle - |1\rangle)/\sqrt{2}$. With respect to this basis, the operator Z takes $|+\rangle$ to $|-\rangle$ and vice versa. We use the states

$|0_L\rangle \equiv |+++ \rangle$ and $|1_L\rangle \equiv |-- - \rangle$ as logical zero and one states for protection against phase flip errors. Error correction is performed as follows:

1. **Encoding:** Encoding is done as Three Qubit Bit Flip Code, then Hadamard Gates are applied to all bits. Circuit for the same is shown below.

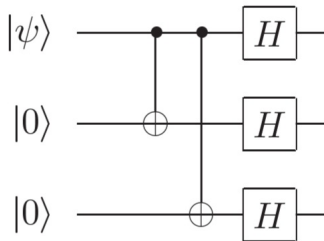


Figure2: Circuit For Encoding Phase Flip Code

2. **Error-Detection or Syndrome Diagnosis:** Using P'_i s of the bit flip code, error syndrome is calculated using operators $H^{\otimes 3}P_iH^{\otimes 3}$ to detect which sign has been flipped. Syndrome Measurements can also be performed using observables $H^{\otimes 3}Z_iZ_jH^{\otimes 3} = X_iX_j$, i.e., comparing the sign of qubits i, j .
3. **Recovery:** operator $HXH = Z$ is applied to the sign i found reversed in syndrome diagnosis. This is followed by H gates being applied to all qubits to recover the original state in standard basis.

4 Constructing Quantum Correcting Codes

We now see construction and working of some formal and more sophisticated quantum error correcting codes and how they work.

4.1 Shor Code

The Shor Code can protect against the effects of an arbitrary error on a single qubit. It was invented by Peter Shor in 1995 [**SHOR**]. The working procedure is described below:

1. Encode the qubit using the phase flip code:

$$a|0\rangle + b|1\rangle \rightarrow a|+++ \rangle + b|-- - \rangle.$$

2. Encode each of these qubits using the bit flip code:

$$|+\rangle \rightarrow (|000\rangle + |111\rangle)\sqrt{2} \text{ and } |-\rangle \rightarrow (|000\rangle - |111\rangle)\sqrt{2}.$$

The resulting nine qubit code is given by:

$$\begin{aligned} |0\rangle &\rightarrow |0_L\rangle \equiv \frac{(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)}{2\sqrt{2}} \\ |1\rangle &\rightarrow |1_L\rangle \equiv \frac{(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)}{2\sqrt{2}}. \\ \implies a|+++\rangle + b|---\rangle &\rightarrow a|0_L\rangle + b|1_L\rangle \end{aligned}$$

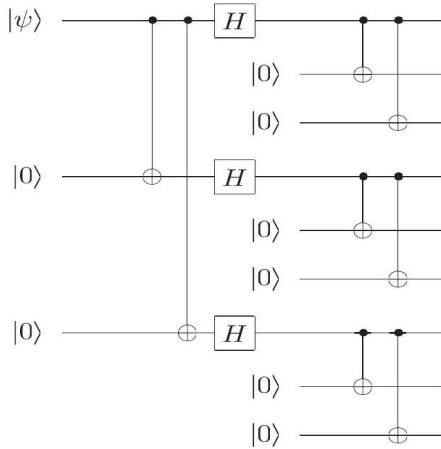


Figure 3. Encoding circuit for the Shor nine qubit code.

3. The stabilizer operators for syndrome measurement of the above code are given by the set below. Subscript indicates the qubit on which error/flip occurs.

$$\begin{aligned} \mathcal{S} = \langle &Z_1Z_2, Z_2Z_3, Z_4Z_5, Z_5Z_6, Z_7Z_8, Z_8Z_9, \\ &X_1X_2X_3X_4X_5X_6, X_4X_5X_6X_7X_8X_9 \rangle \end{aligned}$$

4. Measurement is performed and then correction results are applied for the type of error detected as discussed previously. The procedure for correction when bit and phase flips occur on first qubits are explained below. In a similar way we can detect and recover from the effects of bit or phase flip errors on any of the nine qubits in the code.

Protection Against Bit Flips

Let a bit flip occur on the first qubit. We perform a measurement of X_1X_2 to compare the first two qubits. Difference indicates a bit flip error on the first or second qubit. Next we compare the second and third qubit by using Z_2Z_3 . We find that they are the same, so the second qubit could not have flipped. Now we recover from the error by flipping the first qubit to its original state.

Protection Against Phase Flips

Let a phase flip occurs on the first qubit. This flips the sign of the first block of qubits, changing $|000\rangle + |111\rangle$ to $|000\rangle - |111\rangle$, and vice versa. A phase flip on any of the first three qubits has this same effect, and the error-correction procedure we describe below is same for any.

Syndrome measurement is done by comparing the sign of the first and second blocks of three qubits using Z_iZ_j operators for appropriate i, j . For example, $(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$ has the same sign ($-$) in both blocks of qubits, while $(|000\rangle - |111\rangle)(|000\rangle + |111\rangle)$ has different signs. Then the sign of the second and third blocks of qubits is compared and found out to be the same. Hence we flip the sign in the first block of three qubits back to its original value.

Thus, the Shor code also enables the correction of combined bit and phase flip errors on a single qubit.

4.2 Classical Error Correcting Codes

Theory of Classical Error Correcting Codes is explained in this section. Refer to [2] for more information and theory. Quantum Error Correcting Codes are constructed using similar principles and defined using same terms.

Let $\mathbb{F}_q$ denote the finite field with q elements. We consider the n dimensional vector space $\mathbb{F}_q^n$ whose elements are n -tuples $a = (a_1, \dots, a_n)$ with $a_i \in \mathbb{F}_q$.

Definition 4.1. Let $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n) \in \mathbb{F}_q^n$. The function defined as

$$d(a, b) := |\{i; a_i \neq b_i\}|$$

is called the Hamming distance on $\mathbb{F}_q^n$.

Definition 4.2. The weight of an element $a \in \mathbb{F}_q^n$ is defined as

$$\text{wt}(a) := d(a, 0) = |\{i; a_i \neq 0\}|.$$

Definition 4.3. A code C is a linear subspace of $\mathbb{F}_q^n$. The elements of C are called codewords. Following terms define a code:

1. We call n the length of C and $\dim C$ (as a $\mathbb{F}_q$ -vector space) the dimension of C .
2. An $[n, k]$ code is a code of length n and dimension k . The minimum distance $d(C)$ of a code $C \neq 0$ is defined as

$$d(C) := \min\{d(a, b) \mid a, b \in C \text{ and } a \neq b\} = \min\{wt(c) \mid 0 \neq c \in C\}.$$

3. An $[n, k]$ code with minimum distance d will be referred to as an $[n, k, d]$ code.

Remark 4.4. 1. If $C \subseteq \mathbb{F}_q^n$ is a linear subspace, C is said to be a linear code. Most codes used in practice belong to this class, hence we will only consider linear codes.

2. For a code C with minimum distance $d = d(C)$ we set $t := \lfloor (d-1)/2 \rfloor$ (where $\lfloor x \rfloor$ denotes the integer part of the real number x). Then C is said to be t -error correcting. Clearly if $u \in \mathbb{F}_q^n$ and $d(u, c) \leq t$ for some $c \in C$ then c is the only code-word with $d(u, c) \leq t$.
3. A simple way to describe a specific code C explicitly is to write down a basis of C (as a vector space over $\mathbb{F}_q$).

Definition 4.5. Let C be an $[n, k]$ code over $\mathbb{F}_q$. A generator matrix of C is a $k \times n$ matrix whose rows are a basis of C .

Definition 4.6. The canonical inner product on $\mathbb{F}_q^n$ is defined by

$$\langle a, b \rangle := \sum_{i=1}^n a_i b_i,$$

for $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n) \in \mathbb{F}_q^n$.

Definition 4.7. If $C \subseteq \mathbb{F}_q^n$ is a code then

$$C^\perp := \{u \in \mathbb{F}_q^n \mid \langle u, c \rangle = 0 \text{ for all } c \in C\}$$

is called the dual of C . The code C is self-dual if $C = C^\perp$ (resp. $C \subseteq C^\perp$).

As $C \odot C^\perp = F_q^n$, dual of an $[n, k]$ code is an $[n, n-k]$ code, and $(C^\perp)^\perp = C$. In particular, the dimension of a self-dual code of length n is $n/2$.

Definition 4.8. A generator matrix G of a code C is a matrix whose rows form a basis for C .

Definition 4.9. A generator matrix H of $C^\perp$ is said to be a parity check matrix for C .

Clearly a parity check matrix H of an $[n, k]$ code C is an $(n - k) \times n$ matrix of rank $n - k$, and

$$C = \{u \in \mathbb{F}_q^n \mid H \cdot u^t = 0\}$$

(where u^t denotes the transpose of u). Hence a parity check matrix 'checks' whether a vector $u \in \mathbb{F}_q^n$ is a code-word or not.

Definition 4.10. Two codes $C_1, C_2 \subseteq \mathbb{F}_q^n$ are said to be equivalent if there is a vector $a = (a_1, \dots, a_n) \in (\mathbb{F}_q^\times)^n$ such that $C_2 = a \cdot C_1$; i.e.,

$$C_2 = \{(a_1 c_1, \dots, a_n c_n) \mid (c_1, \dots, c_n) \in C_1\}.$$

One of the basic problems in algebraic coding theory is to constructing codes whose dimension and minimum distance are large in comparison with their length given a fixed alphabet $\mathbb{F}_q$. If the dimension of a code is large (with respect to its length), then its minimum distance is small. The simplest bound is the Singleton Bound, explained below.

Proposition 4.11 (Singleton Bound). For an $[n, k, d]$ code C holds

$$k + d \leq n + 1.$$

Proof. Consider the linear subspace

$$E := \{(a_1, \dots, a_n) \in \mathbb{F}_q^n \mid a_i = 0 \text{ for all } i \geq d\} \subseteq \mathbb{F}_q^n.$$

$\forall a \in E \text{ wt}(a) \leq d - 1 \implies E \cap C = \{0\}$. As $\dim E = d - 1$,

$$\begin{aligned} \implies k + (d - 1) &= \dim C + \dim E \\ &= \dim(C + E) + \dim(C \cap E) = \dim(C + E) \leq n. \end{aligned}$$

□

4.3 CSS Code

Calderbank-Shor-Steane codes, known as *CSS* codes are a subclass of the more general class of stabilizer codes, explained later. Their construction is explained below:

Let C_1 and C_2 be $[n, k_1]$ and $[n, k_2]$ classical linear codes such that $C_2 \subset C_1$ and C_1 and $C_2^\perp$ both correct t errors. We will define an $[n, k_1 - k_2]$ quantum code $\text{CSS}(C_1, C_2)$ capable of correcting errors on t qubits as the *CSS* code of C_1 over C_2 . The construction is given below:-

Let $x \in C_1$ be an arbitrary in the code C_1 . Define the quantum state $|x + C_2\rangle$ as

$$|x + C_2\rangle \equiv \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} |x \oplus y\rangle.$$

Let x' is an element of C_1 such that $x - x' \in C_2 \implies |x + C_2\rangle = |x' + C_2\rangle$. If x and x' belong to different cosets of C_2 , then $|x + C_2\rangle$ and $|x' + C_2\rangle$ are orthonormal states.

The quantum code $\text{CSS}(C_1, C_2)$ is defined to be the vector space spanned by the states $|x + C_2\rangle$ for all $x \in C_1$. The number of cosets of C_2 in C_1 is $|C_1| / |C_2|$ so the dimension of $\text{CSS}(C_1, C_2)$ is $|C_1| / |C_2| = 2^{k_1 - k_2}$, and therefore $\text{CSS}(C_1, C_2)$ is an $[n, k_1 - k_2]$ quantum code.

Correcting Bit And Phase Flip Errors

1. Let the bit flip (phase) errors are described by an n bit vector e_1 with 1 s where bit (phase) flips occurred, and 0s elsewhere. Let $|x + C_2\rangle$ be the original state. Then the corrupted state is:

$$\frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} (-1)^{(x+y) \cdot e_2} |x + y + e_1\rangle.$$

2. To detect where bit flips occurred, introduce an ancilla containing sufficient qubits to store the syndrome for the code C_1 , and initially in the all zero state $|0\rangle$. We apply the parity matrix H_1 for the code C_1 , taking $|x + y + e_1\rangle |0\rangle$ to $|x + y + e_1\rangle |H_1(x + y + e_1)\rangle = |x + y + e\rangle |H_1 e_1\rangle$, since $(x + y) \in C_1$ is annihilated by the parity check matrix. This produces the state:

$$\frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} (-1)^{(x+y) \cdot e_2} |x + y + e_1\rangle |H_1 e_1\rangle$$

3. Error-detection for the bit flip errors is done by measuring the ancilla to obtain the result $H_1 e_1$ and discarding the ancilla, giving the state

$$\frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} (-1)^{(x+y) \cdot e_2} |x + y + e_1\rangle.$$

Knowing the error syndrome $H_1 e_1$ we can find the error e_1 . Recovery is performed simply by applying NOT gates to the qubits at whichever positions in the error e_1 a bit flip occurred.

$$\frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} (-1)^{(x+y) \cdot e_2} |x + y\rangle.$$

4. To detect phase flip errors we apply Hadamard gates to each qubit, taking the state to

$$\frac{1}{\sqrt{|C_2|} 2^n} \sum_z \sum_{y \in C_2} (-1)^{(x+y) \cdot (e_2+z)} |z\rangle,$$

where the sum is over all possible values for n bit z . Setting $z' \equiv z + e_2$, this state becomes:

$$\frac{1}{\sqrt{|C_2|} 2^n} \sum_{z'} \sum_{y \in C_2} (-1)^{(x+y) \cdot z'} |z' + e_2\rangle.$$

If $z' \in C_2^\perp \implies \sum_{y \in C_2} (-1)^{y \cdot z'} = |C_2|$. If $z' \notin C_2^\perp$ then $\sum_{y \in C_2} (-1)^{y \cdot z'} = 0$. Thus the state can be written as:

$$\frac{1}{\sqrt{2^n / |C_2|}} \sum_{z' \in C_2^\perp} (-1)^{x \cdot z'} |z' + e_2\rangle,$$

which is like a bit flip error described by the vector e_2 .

5. For the error-detection for bit flips we introduce an ancilla and reversibly apply the parity check matrix H_2 for $C_2^\perp$ to obtain $H_2 e_2$, and correct the ‘bit flip error’ e_2 , obtaining the state

$$\frac{1}{\sqrt{2^n / |C_2|}} \sum_{z' \in C_2^\perp} (-1)^{x \cdot z'} |z'\rangle.$$

The error-correction is completed by again applying Hadamard gates to each qubit, as done for phase flip code before giving the original encoded state:

$$\frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} |x + y\rangle.$$

4.4 Stabilizer Codes

Consider the EPR state of two qubits

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Observe that $X_1X_2|\psi\rangle = |\psi\rangle$ and $Z_1Z_2|\psi\rangle = |\psi\rangle$; i.e., the state $|\psi\rangle$ is stabilized by the operators X_1X_2 and Z_1Z_2 . The idea of the stabilizer formalism is that many quantum states, errors on qubits, measurements in computational basis and operations are compactly described with the help of operators that stabilize them rather than the state itself. We now study Pauli Group and stabilizer definitions.

Definition 4.12. *The Pauli group consists of all the Pauli matrices and multiplicative factors $\pm 1, \pm i$:*

$$G_1 \equiv \{\pm I, \pm iI, \pm X, \pm iX, \pm Y, \pm iY, \pm Z, \pm iZ\}$$

This set of matrices forms a group under the operation of matrix multiplication.

The general Pauli group on n qubits consists of all n -fold tensor products of Pauli matrices, and multiplicative factors $\pm 1, \pm i$.

Definition 4.13. *Let S be a subgroup of G_n and V_S be the set of n qubit states which are fixed by every element of S . V_S is the vector space stabilized by S , and S is said to be the stabilizer of the space V_S .*

Definition 4.14. *A set of elements $g_1, \dots, g_l$ in a group G is said to generate the group G if every element of G can be written as a product of elements from the list $g_1, \dots, g_l$, and we write $G = \langle g_1, \dots, g_l \rangle$.*

Example 4.15. *For $n = 3$ qubits and $S \equiv \{I, Z_1Z_2, Z_2Z_3, Z_1Z_3\}$. The subspace fixed by Z_1Z_2 is spanned by $|000\rangle, |001\rangle, |110\rangle$ and $|111\rangle$, and the subspace fixed by Z_2Z_3 is spanned by $|000\rangle, |100\rangle, |011\rangle$ and $|111\rangle$. Hence $V_S = \langle |000\rangle, |111\rangle \rangle$ and $S = \langle Z_1Z_2, Z_2Z_3 \rangle$, as $Z_1Z_3 = (Z_1Z_2)(Z_2Z_3)$ and $I = (Z_1Z_2)^2$.*

Constructing Stabilizer Codes

Definition 4.16. *An $[n, k]$ stabilizer code is defined to be the vector space V_S stabilized by a subgroup S of G_n such that $-I \notin S$ and S has $n - k$ independent and commuting generators, $S = \langle g_1, \dots, g_{n-k} \rangle$. We denote this code $C(S)$.*

Suppose $C(S)$ is a stabilizer code corrupted by an error $E \in G_n$. Three possibilities arise:

1. The code space when E anti-commutes with an element of the stabilizer (i.e., $Eg = -gE, s \in S$): Here, E takes $C(S)$ to an orthogonal subspace, and the error can be detected (and perhaps corrected after detection) by performing an appropriate projective measurement.
2. If $E \in S$: The ‘error’ E doesn’t corrupt the space at all.
3. $E \notin S$ but commutes with all the elements of S , i.e., $Eg = gE$ for all $g \in S$: The set of $E \in G_n$ such that $Eg = gE$ for all $g \in S$ is known as the centralizer of S in G_n and is denoted $Z(S)$. In fact, for the stabilizer groups S , the centralizer is identical to the normalizer of S denoted $N(S)$, which consists of all elements E of G_n such that $EgE^\dagger \in S$ for all $g \in S$.

We now define the conditions in errors for which error correction can be performed.

Theorem 4.17. (*Error-correction conditions for stabilizer codes*) *Let S be the stabilizer for a stabilizer code $C(S)$. Suppose $\{E_j\}$ is a set of operators in G_n such that $E_j^\dagger E_k \notin N(S) - S$ for all j and k . Then $\{E_j\}$ is a correctable set of errors for the code $C(S)$.*

Note that set set of errors anti- commutes with set of stabilizers. Refer to Chapter 8 of [1] for proof.

Properties Of Stabilizers

The stabilizers $P_i \in S$ of an $[n, k, d]$ code satisfy the following properties:

1. They must be Pauli-group elements, $P_i \in \mathcal{G}_n$.
2. They must stabilize all logical states $|\psi\rangle_L$ of the code, i.e., $P_i|\psi\rangle_L P_i^\dagger = (+1)|\psi\rangle_L$.
3. All the stabilizers of a code must commute with one another, i.e., $[P_i, P_j] = P_i P_j - P_j P_i = 0 \forall i$ and j . This property is necessary to ensure the stabilizers can be measured independent of their ordering.

The stabilizers P_i of an $[n, k, d]$ code form the Abelian subgroup S of the Pauli Group defined mathematically as below:

$$S = \{P_i \in \mathcal{G}_n | P_i|\psi\rangle_L = (+1)|\psi\rangle_L \forall |\psi\rangle_L \wedge [P_i, P_j] = 0 \forall (i, j)\}.$$

4.4.1 Example: The $[4, 2, 2]$ Detection Code

The $[4, 2, 2]$ detection code is the smallest stabilizer code against X and Z -errors.

A two-qubit register $|\psi\rangle_1|\psi\rangle_2$ is entangled across four qubits to give the code state $|\psi_1\psi_2\rangle_L$. As there are two encoded logical qubits in the $[4, 2, 2]$ code, its codespace is four-dimensional and is spanned by

$$\mathcal{C}_{[4,2,2]} = \text{span} \left\{ \begin{array}{l} |00\rangle_L = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle) \\ |01\rangle_L = \frac{1}{\sqrt{2}}(|0110\rangle + |1001\rangle), \\ |10\rangle_L = \frac{1}{\sqrt{2}}(|1010\rangle + |0101\rangle) \\ |11\rangle_L = \frac{1}{\sqrt{2}}(|1100\rangle + |0011\rangle) \end{array} \right\}.$$

The stabilizers of the above logical basis states are:

$$\mathcal{S}_{[4,2,2]} = \langle X_1X_2X_3X_4, Z_1Z_2Z_3Z_4 \rangle.$$

For the single-qubit X -errors ($E = \{X_1, X_2, X_3, X_4\}$), we see that they all anti-commute with the $Z_1Z_2Z_3Z_4$ stabilizer. The single-qubit Z -errors ($E = \{Z_1, Z_2, Z_3, Z_4\}$) anti-commute with the $X_1X_2X_3X_4$ stabilizer. The syndrome table for the $[4, 2, 2]$ code for all single-qubit X -, Z - and Y -errors is given below.

Error	Syndrome, S	Error	Syndrome, S	Error	Syndrome, S
X_1	10	Z_1	01	Y_1	11
X_2	10	Z_2	01	Y_2	11
X_3	10	Z_3	01	Y_3	11
X_4	10	Z_4	01	Y_4	11

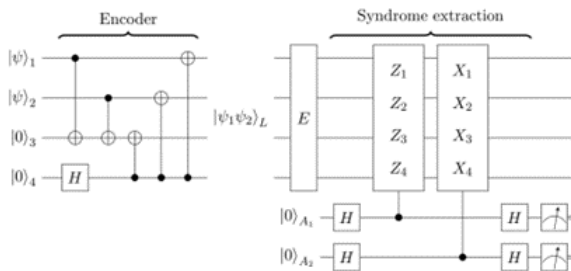


Figure 4. Circuit diagram for the four-qubit code.

The information contained in two-qubit register $|\psi\rangle_1, |\psi\rangle_2$ is distributed across two redundancy qubits, $|0\rangle_3, |0\rangle_4$, to create a logical two bit encoded state $|\psi_1\psi_2\rangle$. The code stabilizers are measured on the code qubits and results copied to the ancilla qubits, whose measurement provides a two-bit error syndrome.

4.4.2 Shor Code

In the computational basis, we get the following codespace for the Shor code

$$\mathcal{C}_{[9,1,3]} = \text{span} \left\{ \begin{array}{l} |0\rangle_9 = \frac{1}{\sqrt{8}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle) \\ |1\rangle_9 = \frac{1}{\sqrt{8}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle) \end{array} \right\},$$

whose stabilizers are given by

$$\mathcal{S}_{[9,3,3]} = \langle Z_1 Z_2, Z_2 Z_3, Z_4 Z_5, Z_5 Z_6, Z_7 Z_8, Z_8 Z_9, \\ X_1 X_2 X_3 X_4 X_5 X_6, X_4 X_5 X_6 X_7 X_8 X_9 \rangle.$$

The syndrome table for single-qubit X - and Z -errors on the nine-qubit code is given below.

Error	Syndrome, S	Error	Syndrome, S
X_1	10000000	Z_1	00000010
X_2	11000000	Z_2	00000010
X_3	01000000	Z_3	00000010
X_4	00100000	Z_4	00000011
X_5	00110000	Z_5	00000011
X_6	00010000	Z_6	00000011
X_7	00001000	Z_7	00000001
X_8	00001100	Z_8	00000001
X_9	00000100	Z_9	00000001

The first six terms are the stabilizers of the bit-flip codes in the three-blocks of the code. The final two stabilizers derive from the stabilizers of the phase-flip code. Each of the X errors produce unique syndromes. In contrast, Z -errors that occur in the same block of the code have the same syndrome. Hence the decoder therefore has insufficient information to differentiate them and outputs the same recovery option. But this does not create a problem, explained below.

Let the decoder's recovery operation be $\mathcal{R} = Z_1$. If the error is $E = Z_1$, the recovery operation trivially restores the logical state. If $E = Z_2$, the recovery operation still restores the logical state as $\mathcal{R}E = Z_1 Z_2 \in$ the stabilizer of $\mathcal{C}_{[9,1,3]}$, and therefore acts on the logical state as follows $Z_1 Z_2 |\psi\rangle_9 = |\psi\rangle_9$. Similarly we can correct all single-qubit errors.

4.4.3 CSS Codes as Stabilizer Codes

Let C_1 and C_2 are $[n, k_1]$ and $[n, k_2]$ classical linear codes such that $C_2 \subset C_1$ and C_1 and $C_2^\perp$ both correct t errors. Define a check matrix:

$$\left[\begin{array}{c|c} H(C_2^\perp) & 0 \\ \hline 0 & H(C_1) \end{array} \right]$$

If this defines a stabilizer code, the check matrix to satisfy the commutativity condition $H(C_2^\perp)H(C_1)^T = 0$. Now,

$$H(C_2^\perp)H(C_1)^T = [H(C_1)G(C_2)]^T = 0$$

as $C_2 \subset C_1$, G is the generator matrix for C_2 as defined in Definition 4.5 in Section 4.2 Classical Error Correcting Codes. This code is exactly $\text{CSS}(C_1, C_2)$, and that it is capable of correcting arbitrary errors on any t qubits.

4.4.4 Matrix Representation of Stabilizer Codes

The $(n - k)$ stabilizers of an $[n, k]$ stabilizer code form the rows of the PCM (Parity Check Matrix) H , which consists of a pair of $(n - k) \times n$ binary matrices H_z and H_x , as given below:

$$H = (H_z \mid H_x)$$

Each row of H corresponds to a stabilizer of $\mathcal{H}$, so that the i^{th} columns of H_z and H_x are used to compute the error imposed on the i^{th} qubit. A binary 1 in the i th column of H_z (or H_x) represents a **Z** (or **X**) Pauli operator in the corresponding stabilizer.

5 Recent Research

This section briefly tells about three candidates for quantum error correcting codes which are actively being researched for implementation currently. This section is mostly non-technical and brief, giving just an overview of these three types of codes.

5.1 Surface Codes

It is challenging to find commuting sets of stabilizers that enable error detection without disturbing the encoded information. Surface codes are built up by ‘patching’ together repeated elements to create a lattice. This ensures easy scaling in size while

ensuring stabilizer commutativity. Surface codes infact can be given a compact pictorial representation- explained by the following example. Figure 5(a) which shows a surface code four-cycle.

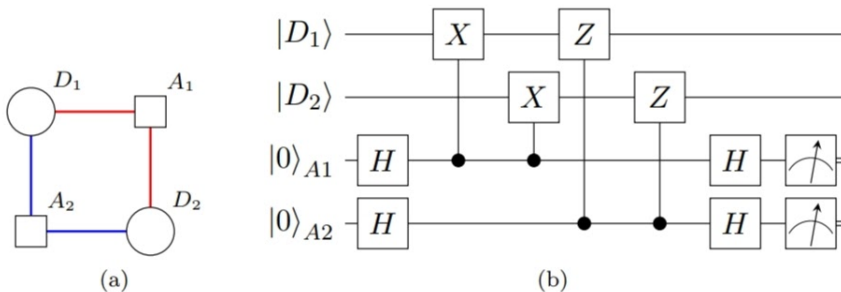


Figure 5. Surface Code 4 Cycle

1. The circles represent the code qubits and the squares the ancilla qubits.
2. The red edges represent controlled- X gates, each controlled on an ancilla qubit A and acting on a data qubit D . The blue edges represent controlled- Z operations, each controlled by an ancilla qubit and acting on a data qubit. These controlled operations are the gates with which the stabilizers of the four-cycle are measured.
3. Ancilla qubit A_1 connects to data qubits D_1 and D_2 via red edges, and therefore measures the stabilizer $X_{D_1}X_{D_2}$. Similarly, ancilla qubit A_2 measures the stabilizer $Z_{D_1}Z_{D_2}$. The stabilizers ($X_{D_1}X_{D_2}$ and $Z_{D_1}Z_{D_2}$ in this case) which commute with one another intersect non-trivially on an even number of code qubits.

The five-qubit surface code is formed by tiling together four four-cycles in a square lattice is shown in Figure 6.

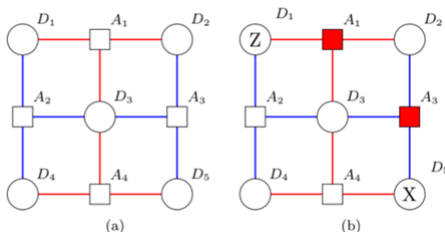


Figure 6.

By inspecting, the stabilizers of the code are found to be :

$$S_{(5,1,2)} = \{X_{D_1}X_{D_1}X_{D_y}, Z_{D_1}Z_{D_1}Z_{D_n}, Z_{D_3}Z_{D_1}Z_{D_n}, X_{D_1}X_{D_1}X_D\}$$

Figure 6(b) shows two examples of error detection on the surface code. The Z_{D_1} -error on qubit D_1 anti-commutes with the $X_{D_1}X_{D_2}X_{D_3}$ stabilizer, and therefore triggers a ‘1’ syndrome- shown by red in the ancilla qubit A_1 . The X_{D_5} -error anti-commutes with the $Z_{D_2}Z_{D_3}Z_{D_5}$ stabilizer and triggers a ‘1’ syndrome measurement in ancilla qubit A_4 . For more details on surface codes, see [P1],[SC].

5.2 LDPC Codes

Gallager in his PhD thesis [LDPC] defined a low density parity check (LDPC) code as one which has a parity check matrix with constant row and column sparsity². LDPC codes are codes specified by a matrix containing mostly 0’s and relatively few 1’s. They do not optimize the error-decoding probability of for a given block length, but provide a simple and efficient decoding scheme, which quite compensates for their disadvantages.

We may divide the QLDPC codes into three main categories on the basis of the Parity Check Matrix H : Calderbank-Shor-Steane (CSS) codes, non-CSS codes and Entanglement-Assisted (EA) codes, as summarized in Figure 7. The CSS-type constructions may also be classified as dual containing and as non-dual-containing codes. For more details about working of these codes, refer [LDPC], [QLDPC] and [QLDPC-1].

²Sparsity of a matrix is the number of zero-valued elements divided by the total number of elements

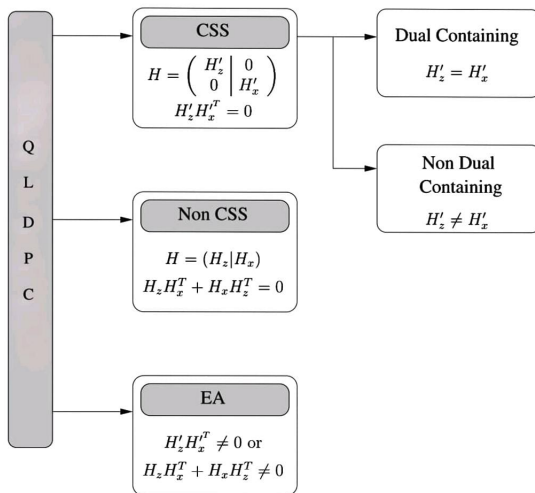


Figure 7. Types of QLDPC Codes

Most of the interest in QLDPC quantum was due to Daniel Gottesman’s result in 2013 [**QLDPC-3**] showing that quantum LDPC codes with a constant encoding rate can reduce the overhead³ of fault-tolerant quantum computation⁴ to a constant. In other quantum fault tolerance schemes, performing longer error free computation requires larger codes and thus increases number of physical qubits. Although, currently Surface codes are considered the leader in quantum error correcting codes, in three long-term QLDPC codes may well turn out to be better for the implementation of quantum computers.[**QLDPC-2**]

5.3 Holographic Error Correcting Codes

Albert Einstein’s general theory of relativity defines gravity as the fabric of space and time (or “space-time”) bending around massive objects. But physicists believe gravity must have a quantum origin from which space-time fabric somehow emerges. In 2014, three young quantum gravity researchers- Ahmed Almheiri, Xi Dong and Daniel Harlow were working in a toy theoretical physics universe called “anti-de Sitter space”. This space works like a hologram- the bendy fabric of space-time in the interior of the universe is a projection that emerges from entangled quantum particles living on its outer boundary. They conjectured in the Journal of High Energy Physics that space-time itself is a quantum error correcting code in anti-de Sitter

³the ratio of physical qubits to logical qubits

⁴A quantum circuit is fault-tolerant against t failures if failures in t elements results in at most t errors per code block (group of qubits corrected together). See [**FAULT**] for more information

(AdS) universes, at least[**ADC-CFT**]. More new quantum error-correcting codes have been discovered that capture more properties of space-time. Read [**QSPT**] for more.

Quantum Gravity is a field of theoretical physics seeking to describe gravity via laws of quantum mechanics and Quantum field theory (QFT) is a theoretical framework that combines classical field theory, special relativity, and quantum mechanics. The anti-de Sitter(AdS)/conformal field theory (CFT) Correspondence is a conjectured relationship between these two kinds of physical theories- it tells an exact duality between quantum gravity on a $(d + 1)$ - dimensional asymptotically-AdS space and a d -dimensional CFT defined on its boundary. Quantum states can be understood in the form of tensor products. A class of tensors called perfect tensors are associated with pure quantum states of many spins such that the entanglement is maximal across any partition of the spins into two sets of equal size. Networks of perfect tensors can be used to construct holographic states and error correcting code. These codes have properties similar of the AdS/CFT correspondence. Read [**HLOQC**] to know more.

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